Hydrogen storage in TiFe-based alloys doped with Cr and S: thermodynamic and structural peculiarities

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Abstract

For the first time the Tian-Calvet differential heat-conducting microcalorimetry has been applied to thermochemical analysis of hydrogen absorption and desorption processes in $TiFe_{0.92}Cr_{0.08}$-H_2 and $TiFe_{0.9}Cr_{0.08}S_{0.02}$-$H_2$ systems. For both studied alloys, the effect of leveling the enthalpy values corresponding to α↔β and β↔γ phase transitions was revealed. As it was found, the alloying of TiFe with chromium and sulfur makes it possible to eliminate the stage of high-temperature activation due to the presence of catalytically active $Ti(Fe,Cr)_2$ phase and S-containing precipitations at grain boundaries. Concurrent substitution of Fe with Cr and S results in a decrease in hydrogen sorption and pressure of the second plateau compared to the substitution with Cr alone. The sulfur-containing alloy exhibited reduced hydrogen storage capacity (1.3 wt.% vs. 1.9 wt.%) due to sulfur occupying interstitial positions in the intermetallic lattice and thus block them for hydrogen atoms. The studied method of composition modification can be considered as a promising approach to improvement of hydrogen storage performance of TiFe-based alloys.

Keywords: Hydrogen storage, TiFe alloys, Thermodynamic parameters, Tian-Calvet calorimeter.

1. Introduction

The development of efficient and safe hydrogen storage systems is a key step towards establishing sustainable energy systems. Among the promising hydride-forming alloys, special attention should be paid to the intermetallic compound TiFe and its alloy derivatives, which combine an attractive storage capacity at low cost. The hydrogen capacity in the TiFe alloy can vary from 1 wt.% to 1.8 wt.% hydrogen and the stability of the stored hydrogen amount is maintained for up to 3000 sorption/desorption cycles. [1,2]. However, TiFe-based alloys produced by traditional methods (i.e. arc-melting) have an important problem associated with the slow and complex activation procedure required to initiate the hydrogenation process. The low hydrogen absorption capacity of TiFe at room temperature can be attributed to the presence of a naturally occurring passivating layer on its surface. This oxide layer acts as a significant barrier to hydrogen diffusion, preventing the dissociation of H_2 molecules and the subsequent penetration of atomic hydrogen into the bulk material. As a result, special treatment conditions must be employed to either disrupt this protective surface layer or prevent its formation [3]. Initial TiFe samples are typically subjected to high-temperature treatment at 400°C in order to achieve this goal.

The number of papers [4-21] demonstrates various improvements in hydrogen performance through doping of TiFe alloy with different metals (Mg, Cr, S, Ni, Mn, V, Al, Cu, Y, Co) without reducing the hydrogen sorption capacity. Substitution metals can be divided into two groups: A elements that substitute Ti (Y, Zr, V, Nb) and B – Fe (Co, Ni, V, Al, Cu, Ni, Cr, S, Mg, Mn). A elements commonly used to tune the strength of metal and hydrogen bonds, while B elements – to improve the electrocatalytic activity and the ability of the alloy to retain hydrogen.

Partially substitution Fe with Mg [4], S [2,4,5], Cu [6,7], Cr [8,10,11,19-21], Mn [6,7,9-11,20,21], Co [13,14], Ni [12,18], Al [15,19], Si [2] leads to improving activation rate, reducing hysteresis, increasing hydrogen capacity, decreasing plateau pressure, and improving cycling properties. However, high concentration of these elements results in significant reduction in hydrogen storage capacity, decrease in cycle stability, and increase in hysteresis.

According to [7], the concurrent doping of TiFe-based alloys with Mn and Cu is suitable for hydrogen storage under low pressures. In work [4], it was shown that the introduction of 1 wt.% sulfur leads to extreme simplification of the hydrogen activation process. The addition of

1-5 wt.% Cr to the TiFe alloy results in increased hardness and brittleness, making it easier to pulverize. This may be the reason for improved hydrogen kinetics and reduced sorption hysteresis [10].

However, modifying TiFe with additional components may lead to the formation of new phases. For instance, Suzuki et al. [5] demonstrated the formation of a network-like Ti_2S phase (Pnnm crystal lattice), which prevent the pulverization of the alloy and serves as catalyst to accelerate and simplify activation, yet it reduces the amount of remaining hydrogen. In work [20] $TiCr_2$ (C14 Laves phase) and $TiMn_{1.5}$ (η phase) were observed via XRD technique. A small amount of Mn makes it possible to obtain a single-phase TiFe-based alloy, while even a slight amount of Cr leads to the formation of an additional phase. Both elements boost the hydrogen absorption rate connecting it with the second phase formation. Dispersed distribution of the second phase along the matrix provides a preferred path for hydrogen and its huge cell volume extension take part. In their next work [21] only $TiCr_2$ phase was obtained. It was demonstrated that increasing Cr amount increment the phase amount, lower the equilibrium pressures, and accelerates the activation without losing the hydrogenation kinetics in the subsequent cycles. Park et al. [11] obtained similar results yet provided huge discussion on hydrogenation process. Improved hydrogen kinetics of TiFe-based alloys with second $TiCr_2$ phase owes to generation of cracks induced by $TiCr_2$ crystal lattice volume expansion. Moreover, simultaneous substitution Fe by two elements results both in improved kinetics at room temperature and discards the need of activation. Also, the $TiFe_2$ phase (C14 Laves phase) can be observed in TiFe-based alloys, yet it does not interact with hydrogen and its appearance decreases hydrogen storage capacity [22-24]. In [25] authors explained improvement of the initial hydrogenation kinetics by the formation of the $TiFe_2$ phase connecting it with the sample embrittlement. However, Jain et al. [18] reports only degradation of hydrogen sorption performance in the presence of the $TiFe_2$ phase.

It is of interest to investigate the interaction of H_2 with an intermetallic compound doped with both Cr and S simultaneously. It is also necessary to evaluate the effect of S on hydride formation, for which an experiment is proposed under the same conditions using a TiFe-based alloy doped with only Cr. To analyze the thermodynamic behavior of substituted alloys, the Tian-Calvet method of differential heat-conducting microcalorimetry was previously successfully applied [14, 26]. This technique makes it possible to determine the hydrogen absorption and desorption enthalpies through direct measurement of the heat reaction and to identify the features of related phase transformations.

Thus, the aim of this study is to investigate the thermodynamics properties and structure changes during hydrogenation in the $TiFe_{0.92}Cr_{0.08}$-H_2 and the $TiFe_{0.9}Cr_{0.08}S_{0.02}$-$H_2$ systems.

2. Materials and Methods

2.1 Alloy preparation

The chips of Ti (purity 99.4%), Fe (purity 99.5%), Cr (purity 99.4%), and iron sulfide FeS (purity 99%) were used for alloy preparation. The ingots of the alloys were fabricated by arc melting the stoichiometric mixtures of the raw materials in an argon atmosphere purified by a Ti getter for residual gas removal. Upon melting, the ingots were turned over and re-melted five times to ensure compositional homogeneity. Obtained alloys $TiFe_{0.92}Cr_{0.08}$ and $TiFe_{0.9}Cr_{0.08}S_{0.02}$ are designated further in the text as TiFeCr and TiFeCrS for simplicity.

2.2 Structure analysis

To investigate the structure of the produced hydrides, the samples were quenched using liquid nitrogen under hydrogen pressure and subsequently maintained at the same temperature (77 K) in air to passivate the surface and prevent decomposition of unstable hydride phases under standard conditions.

The structure and phase composition of the obtained materials were analyzed on a DRON-type X-ray diffractometer using CoKα radiation with carbon monochromator. Diffractograms were acquired in 20-130° 2θ range with a step size of 0.1° and exposition time of 3 seconds, a sample rotating, a voltage of 40 kV and a current of 20 mA. Rietveld refinement was used for qualitative and quantitative phase analysis, crystal lattice parameters refinement, and preferred orientation.

The sample microstructure was examined by scanning electron microscopy (SEM) using a TESCAN VEGA COMPACT (accelerating voltage 20 kV) equipped with an Oxford energy dispersive X-ray spectrometer (EDS).

2.3 Hydrogenation experiments

Investigation of hydrogen interaction with the alloy was performed on an original measuring device consisting of a Tian-Calvet heat-conducting microcalorimeter (DAK-1a) connected to a Siverts-type system for gas supply. A metal hydride unit (based on $LaNi_5$) was used as a source of high purity hydrogen (99.999%). Calorimetric measurements were carried out in isothermal mode at 308 K; the temperature in the calorimeter was maintained with an accuracy of 0.2 K. The design of the experimental setup was described in detail in [26, 27].

Prior to the measurements, the reactor with the tested sample was degassed to a residual pressure of 0.5 Pa at room temperature. No activation was required for the initial interaction with hydrogen. PC isotherms were measured after three hydrogen absorption-desorption cycles to ensure full activation and reproducibility of PC curves. Hydrogenation and dehydrogenation were carried out at room temperature, and between measurements the samples were additionally heated to 523 K to completely remove hydrogen.

During absorption measurements hydrogen was supplied to the system in portions of 0.0002–0.001 mol H_2 with simultaneous recording of the differential signal of the calorimeter thermoelectric element, which refers to the heat flow resulting from the reaction. Integrating the heat flow over the time gives thus the total amount of the heat released during a certain reaction stage. The amount of absorbed gas was defined from the pressure change in a calibrated volume. The hydride composition was determined with an accuracy of ±0.05 wt%.

To evaluate the desorption behavior, hydrogen was released in portions from the sample into a pre-evacuated calibrated volume with registration of the related thermal effect. The calorimetric data were collected from three independent measurement series.

3. Results and Discussion

3.1 Structure characterization

Structure analysis was conducted using powder X-ray diffraction, which revealed a strongly pronounced crystallographic texture of the (110) line in both alloys. Such planes are close-packed in B2-type structures and have low surface energy [28]; thus, they should be preferable for hydrogen occupation. Obtaining this crystallographic texture exposes overlapping tetrahedral [Ti2Fe2] and octahedral [Ti2Fe4] and [Ti4Fe2] interstices. Machlin [29] presents calculations supporting suggestion that hydrogen prefer tetrahedral interstices yet notes that the relaxation caused by hydrogen may lead to a change in the preferred interstice type to the octahedral ones. According to calculations in [30], the radius of the [Ti4Fe2] octahedral interstice is almost twice as large as that for [Ti2Fe4] and quite close to the covalent radius of H (~0.32Å), while the hydrogen absorption energy in the [Ti2Fe4] interstice is up to 1 eV higher than in the [Ti4Fe2] interstice. This assumption is validated by neutron diffraction experiments conducted in [31], where deuterium atoms occupy the [Ti4Fe2] octahedral interstices. Moreover, in the physisorption process, dissociated molecular hydrogen prefers to occupy [Ti-Ti-Fe] sites formed by [Ti4Fe2] interstices, while in the chemisorption process it occupies [Ti-Ti-Fe] sites formed by [Ti2Fe4] interstices [32,33].

According to [34], the relation between (110)α and (100)β is coherent and is slightly susceptible to volumetric expansion, and the dominant interface appear parallel to the (110)α plane. However, from the perspective of chemical contribution, the most energetically stable interface is (001)α‖(001)β, but only at the initial hydrogenation stage. During the hydrogenation process, expansion of crystal lattice occurs, and the boundaries between the initial and hydride phases tend to become less coherent due to the formation of a dislocations network. This helps to accommodate the significant increase in volume, thereby reducing the strain energy associated with coherence. It should be noted that β-hydride can be obtained from doubled TiFe crystal lattice which has tetragonal extension along [110] direction and following orthorhombic distortion of lattice parameters. Thus, the (110) crystallographic texture might be preferred for the phase transition and could facilitate faster hydrogen absorption.

XRD analysis demonstrates the formation of two-phase alloys (Fig. 1 a,b). The major component in both alloys (80%) was identified as a body-centered cubic (BCC) phase. According to EDX analysis (Table 1 and Fig. 2), it refers to Ti(Fe,Cr) with chromium concentration close to 2 at.%. The minor phase (20%) crystallizes in the $P6_3/mmc$ space group (C14 Laves phase) and can be described as $Ti(Fe,Cr)_2$ with Cr content close to 7 at.%. Crystal lattice parameters of TiFe-type phase were a=0.2987 nm for TiFeCr alloy and a=0.2980 nm for TiFeCrS alloy. This is somewhat higher than the corresponding values for individual intermetallic compound TiFe and is consistent with the larger atomic radius of chromium (R_{Cr} = 0.130 nm) as a substitute for iron (R_{Fe} = 0.126 nm) [19,35,36]. It is worth noting that in the four-component alloy sulfur is present as a minor addition in the TiFe phase (about 3 at.%) and is mainly located along grain boundaries (Fig. 3) in the form of titanium compounds, which is not detected by XRD due to the low concentration of this phase.

As it was pointed in [11,20,21], formation of the chromium containing Laves phase distributed along the TiFe matrix improves hydrogen sorption kinetic: it acts as a catalyst and facilitates the penetration of hydrogen into the sample to a greater depth. The sulfur distribution is similar to described previously in [5] and it also should enhance hydrogen sorption kinetic.

Table 1 – Chemical composition of the main phases in the studied alloys

Alloy	Ti(Fe,Cr)				$Ti(Fe,Cr)_2$			
	Ti, at.%	Fe, at.%	Cr, at.%	S, at.%	Ti, at.%	Fe, at.%	Cr, at.%	S, at.%
TiFeCr	51.1	46.1	2.8	--	41.0	52.3	6.7	-
TiFeCrS	52.2	41.9	2.8	3.1	40.6	52.8	6.6	0

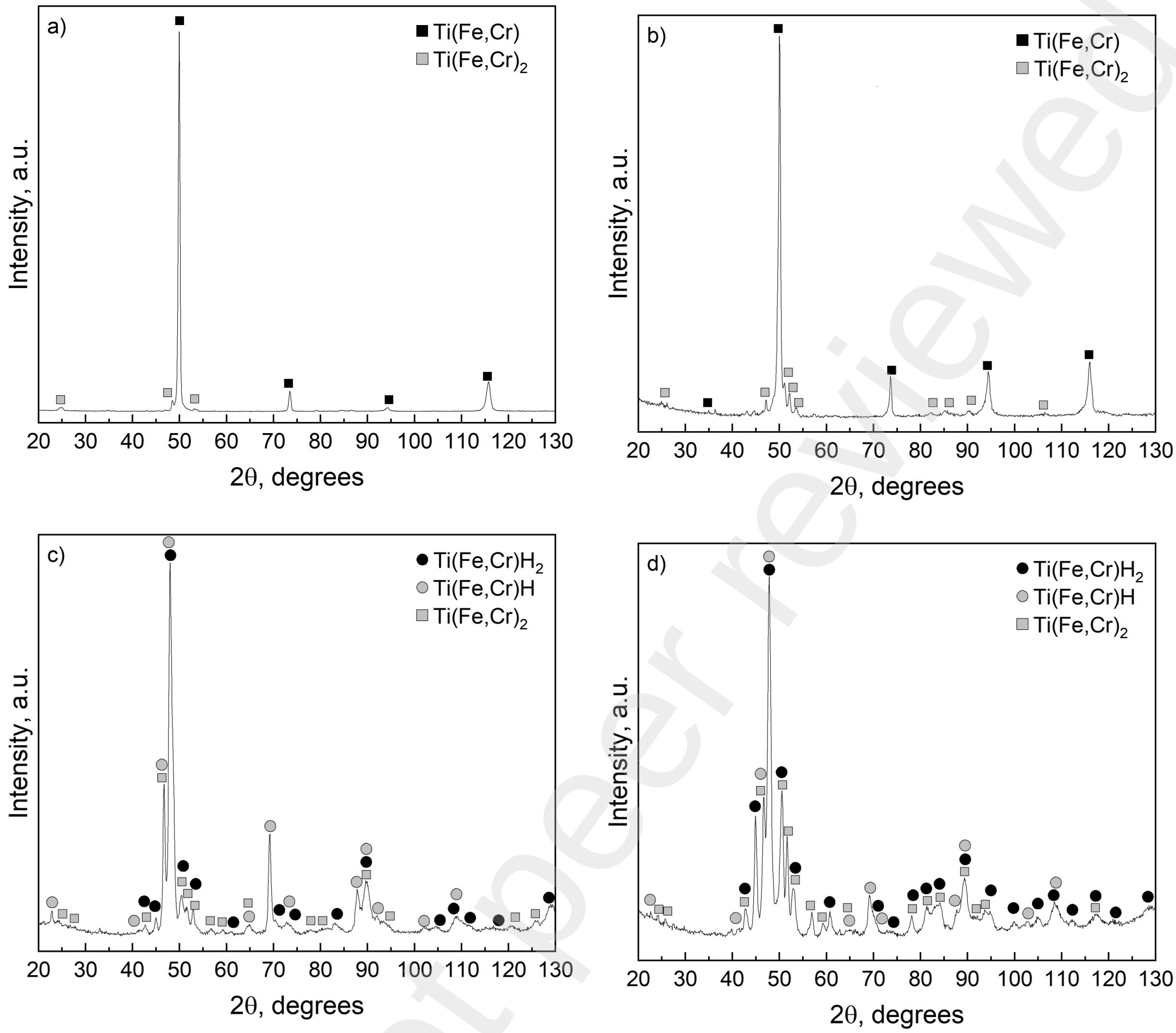
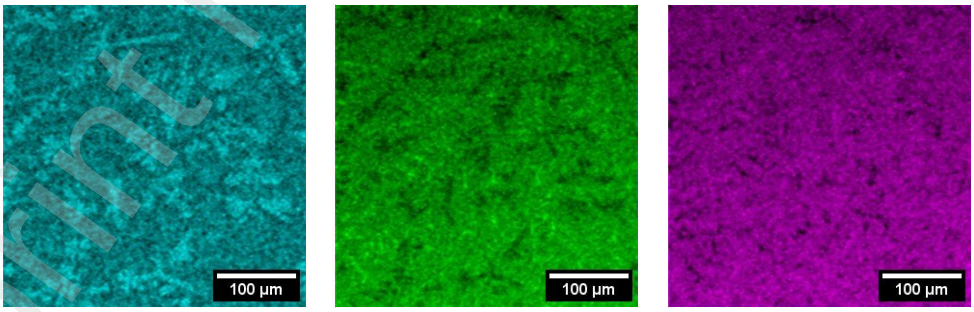
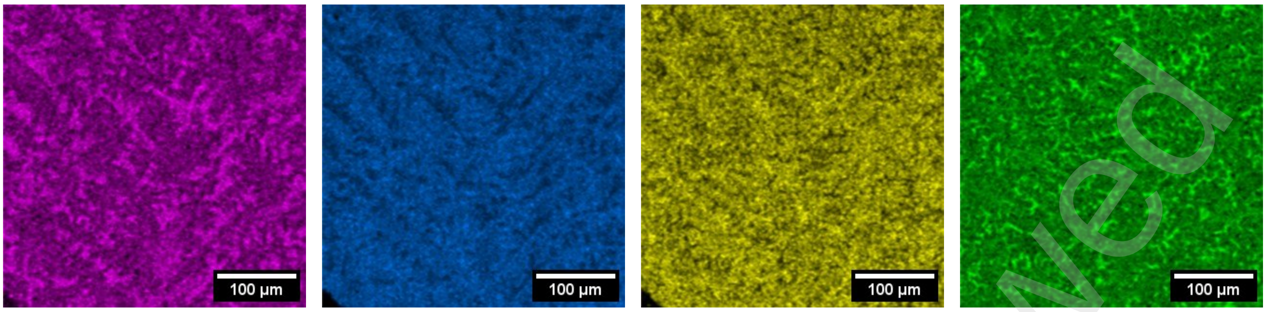


Fig. 1 – XRD patterns of the initial TiFeCr (a) and TiFeCrS (b) alloys and their hydrogenation products (c,d)



a) b) c)

Fig. 2 – Element mapping of as-cast TiFeCr alloy: Cr (a), Ti (b), Fe (c)



a) b) c) d)

Fig. 3 – Element mapping of as-cast TiFeCrS alloy: Cr (a), Ti (b), Fe (c), and S (d)

During hydrogenation (Fig. 1 c,d), in both alloys Ti(Fe,Cr) phase transforms into β-hydride and γ-hydride, while the hydrogenated form of $Ti(Fe,Cr)_2$ Laves phase was not observed (only a solid solution with a small amount of hydrogen according to unit cell parameters). The unit cell volume of the β- and γ-hydrides in TiFeCr and TiFeCrS alloys was bound to be somewhat larger (Table 2) than the literature values for the pure TiFe alloy [35,36]. This can be explained by the presence of alloying elements, Cr and S.

Table 2 – Phase composition and structure parameters of TiFe-based alloys and their hydrides

Alloy	State	Space group	Unit cell parameters, nm	Phase content, %	V, nm^3	ΔV/V0, %
TiFe [5]	Initial	Pm3m	a=0.2976	100	0.0264	-
	Hydrogenation	$P222_1$	a=0.2956 b=0.4543 c=0.4388	100*	0.0589	11.6
		Cmmm	a=0.7029 b=0.6233 c=0.2835	100*	0.1242	17.6
TiFeCr	Initial	Pm3m	a=0.2987(3)	80(5)	0.0267	-
		$P6_3/mmc$	a=0.4855(8) c=0.7877(9)	20(5)	0.1608	-
	Hydrogenation	$P222_1$	a=0.3055(1) b=0.4508(2) c=0.4401(2)	70(5)	0.0606	13.5
		Cmmm	a=0.7047(4) b=0.6265(4)	10(5)	0.1252	17.2

			c=0.2835(2)			
		$P6_3/mmc$	a=0.4924(8) c=0.8054(8)	20(5)	0.1643	2.2
TiFeCrS	Initial	TiFe	a=0.2980(2)	80(5)	0.0265	-
		$P6_3/mmc$	a=0.4853(4) c=0.7918(8)	20(5)	0.1615	-
	Hydrogenation	$P222_1$	a=0.3081(2) b=0.4509(3) c=0.4397(2)	50(5)	0.0609	14.9
		Cmmm	a=0.7060(3) b=0.6244(4) c=0.2838(3)	30(5)	0.1251	18
		$P6_3/mmc$	a=0.4903(8) c=0.8006(4)	20(5)	0.1667	3.2

* 100% either monohydride or dihydride is formed

3.2 Calorimetric analysis of hydrogen interaction with alloys

TiFeCr and TiFeCrS alloys reacted with hydrogen at room temperature without high-temperature activation. As expected, modifying TiFe alloys by Cr and S simplified activation process. Such improvement can be attributed to the formation of the $Ti(FeCr)_2$ Laves phase in both alloys [11, 20,21]. Besides, as it was reported in [2, 5], the introduction of a small amount of sulfur prevents the pulverization of the alloy, due to the formation of a network structure of Ti_2S as a secondary phase at grain boundaries and can also increase resistance to poisoning.

Two two-phase regions, namely α+β and β+γ, were identified in both alloys. The sulfur-containing alloy has a lower hydrogen storage capacity. The amount of hydrogen stored in the TiFeCrS alloy does not exceed 1.3 wt.%, whereas that of the TiFeCr alloy is 1.9 wt.%. This is likely due to sulfur atoms, which occupy some interstitial positions in the intermetallic lattice and thus block them for hydrogen atoms. Fig. 4 demonstrates that the α+β equilibrium hydrogen pressure for both alloys decreased with respect to pure TiFe. Moreover, this effect is more pronounced for TiFe alloyed with Cr rather than with both Cr and S.

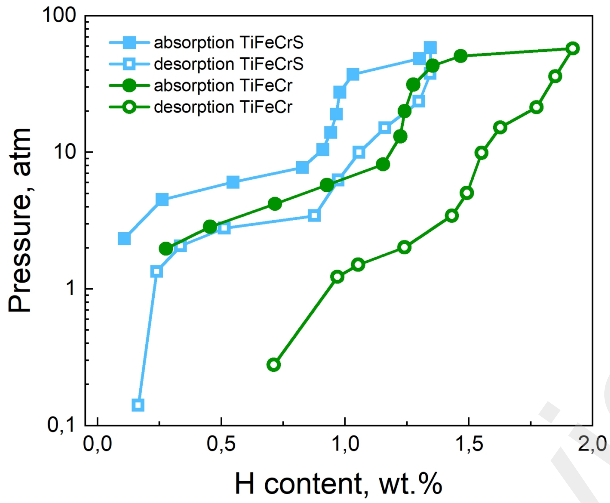


Fig. 4 – PC-isotherms for TiFeCrS (blue line and squares) and TiFeCr (green line and circles) alloys at 308 K.

The method of direct experimental determination of the differential molar enthalpy of hydrogen absorption/desorption (Fig. 5-6) provides much more accurate values for ΔH of processes occurring in a system than those obtained using the Van't-Hoff equation, which relies mainly on the PCT method. The thermodynamic parameters calculated using the Van't-Hoff equation depend strongly on the correct definition of phase boundaries and on the choice of the plateau midpoint. For example, in the case of $\beta \leftrightarrow \gamma$ phase transition the corresponding plateau has a significant slope, which complicates the calculations. The calorimetric titration technique characterizes the system under study at each point of the PCT diagram. The values for the heat of the reaction are strictly related to the composition of the hydride. In addition, the experiment can be conducted at a single temperature (308 K in this case). Therefore, when using the Tian-Calvet microcalorimetry method, a subjective assessment of the equilibrium pressure value "at the middle of the plateau" does not affect the result. Table 3 represent the values of thermodynamic characteristics obtained using the Tian-Calvet microcalorimeter.

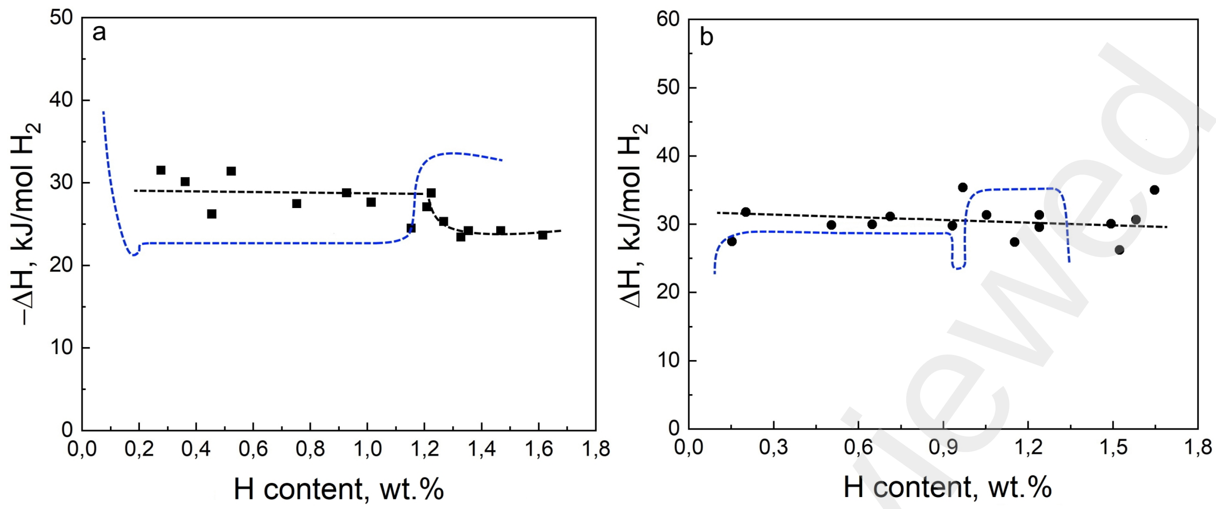


Fig. 5 – Dependence of the differential molar enthalpy absorption (a) and desorption (b) on the hydrogen content in TiFeCr alloy. The blue dotted line corresponds to the data on TiFe from [26]

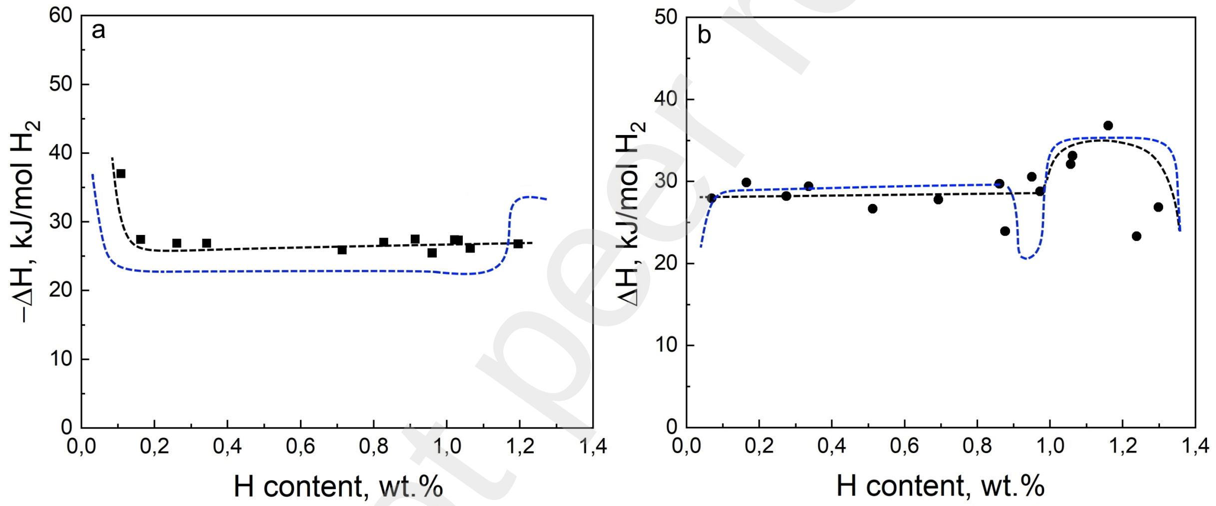


Fig. 6 – Dependence of the differential molar enthalpy absorption (a)/desorption (b) on the hydrogen content in TiFeCrS alloy. The blue dotted line corresponds to the data on TiFe from [26]

Table 3 – Thermodynamic properties of TiFe-based alloys

Alloy composition	Technique	Phase transition	ΔH_{abs}, kJ/mol H_2	ΔH_{des}, kJ/mol H_2	References
$TiFe_{0.92}Cr_{0.08}$	Calorimetry 308K	α↔β	-28.4±1.6	30.0±1.5	This work
		β↔γ	-24.9±0.8	30.2±1.6	
$TiFe_{0.9}Cr_{0.08}S_{0.02}$	Calorimetry 308K	α↔β	-26.4±1.1	28.8±1.2	This work
		β↔γ	-26.9±0.6	32.1±2.4	
TiFe	Calorimetry 308K	α↔β	-23.0±1.1	27.2±0.8	26
		β↔γ	-33.9±0.5	35.2±2.5	

TiFe	Calorimetry 314K	$\alpha\leftrightarrow\beta$	-23.4±0.8	23.4±0.8	37
		$\beta\leftrightarrow\gamma$	-27.8±1.0	30.2±1.8	
TiFe	Van't-Hoff 274-344 K	$\alpha\leftrightarrow\beta$	-25.4±1.6	25.6±1.0	37
		$\beta\leftrightarrow\gamma$	-32.2±4.2	38.8±4.0	
$Ti_{1.05}FeS_{0.02}$	Van't-Hoff 298-353 K	$\alpha\leftrightarrow\beta$	-21.8	-	5
$TiFe_{0.9}Cr_{0.1}$	Van't-Hoff 283-323 K	$\alpha\leftrightarrow\beta$	-27.2	34.2	10
$TiFe_{0.8}Cr_{0.2}$	Van't-Hoff 303-343 K	$\alpha\leftrightarrow\beta$	-35.39±1.00	36.34±1.99	11
TiFe	Van't-Hoff 278-357 K	$\alpha\leftrightarrow\beta$	-21.8	27.6	38
$TiFe_{0.9}Cr_{0.1}$	Van't-Hoff 278-357 K	$\alpha\leftrightarrow\beta$	-	30.1	
$TiFe_{0.8}Cr_{0.2}$	Van't-Hoff 278-357 K	$\alpha\leftrightarrow\beta$	-33.1	35.6	

Alloying TiFe with Cr improves stability of monohydride and lower dissociation hydrogen pressure. However, the addition of sulfur destabilizes the hydride phase and leads to an increase in pressure on the plateau.

It is worth noting that there is a known peculiar pattern for TiFe-H_2 system [26, 37]. In contrast to conventional metal hydride systems for TiFe the absolute value of the enthalpy change at the second plateau ($\beta\leftrightarrow\gamma$ transition) is higher compared to the first one ($\alpha\leftrightarrow\beta$). The substituted alloys studied in the present work break this previously observed feature: during hydrogenation, the absolute enthalpy values for second plateau are approximately equal (TiFeCrS) or even lower (TiFeCr) than for the first one. Such an unexpected effect is likely due to the difference in location of hydrogen atoms in β and γ phases. According to previous structural investigations [1,39,40], ~90% of the hydrogen atoms occupies all [Ti4Fe2] octahedral sites in monohydride, yet all [Ti4Fe2] and ~91% of the [Ti2Fe4] octahedral interstices in dihydride. According to Ko et al. [41] Cr and S atoms prefer to Fe site. DTF calculations demonstrate that doping pure TiFe with Cr and S lead to decrease in the formation energy of hydride from -26.09 kJ/mol H_2 to -40.63 kJ/mol H_2 and -40.47 kJ/mol H_2 respectively. Additionally, [Ti2Fe4] interstices are occupied last because their size is relatively small compared to [Ti4Fe2] interstices and energetically unfavorable. When Cr and S atoms are introduced, the substitution of iron atoms most likely occurs in these same octahedral interstices containing 4Fe atoms. Sulfur and chromium atoms are larger than iron atoms, and their substitution should increase the size of these interstices, which might influence hydrogen-metal bonding energy and promote reduction of the energy required for hydrogen incorporation into these sites. This facilitates the β↔γ phase transition that we observe in thermochemical analysis.

Conclusions

Using the differential heat-conducting Tian-Calvet microcalorimetry, the dependence of the reaction enthalpy on hydrogen content in solid phase was evaluated for the first time for $TiFe_{0.92}Cr_{0.08}$-H_2 and $TiFe_{0.9}Cr_{0.08}S_{0.02}$-$H_2$ systems. It was shown that the hydrogenation process of both alloys does not require high temperature activation. Concurrent substitution of Fe with Cr and S results in a slight decrease in hydrogen capacity and pressure of the second plateau but an increase in the pressure of the first plateau compared to the substitution with Cr alone. This is likely due to the presence of sulfur atoms surrounding the interstitial positions occupied by hydrogen in the lattice. The crystal structure analysis revealed the formation of $Ti(Fe,Cr)_2$ phase in both alloys. This phase might act as a catalyst and facilitates the penetration of hydrogen into the sample to a greater depth. During hydrogenation, the main TiFe phase is successively transformed into the into β-hydride and γ-hydride.

The enthalpy values for the first and second plateaus of hydrogen absorption in TiFeCr were evaluated as -28.4 kJ/mol H_2 and -24.9 kJ/mol H_2, respectively. TiFeCrS alloy demonstrates the equalization of the plateau enthalpies (-26.4 kJ/mol H_2 and -26.9 kJ/mol H_2, respectively). As expected, the corresponding parameters of the hydrogen desorption are noticeably higher that correlates with the pronounced absorption/desorption hysteresis.

Credit authorship contribution statement

A. Korol: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation; V. Zadorozhnyy and S. Klyamkin: Writing – review & editing, Validation, Methodology, Supervision, Founding acquisition; M. Zadorozhnyy E. Berdonosova, Ermakov P.D.: Investigation, Formal analysis, Methodology, Data curation. A. Bazlov: Methodology, Resources, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The research was supported by the Russian Science Foundation, project №24-22-00246.

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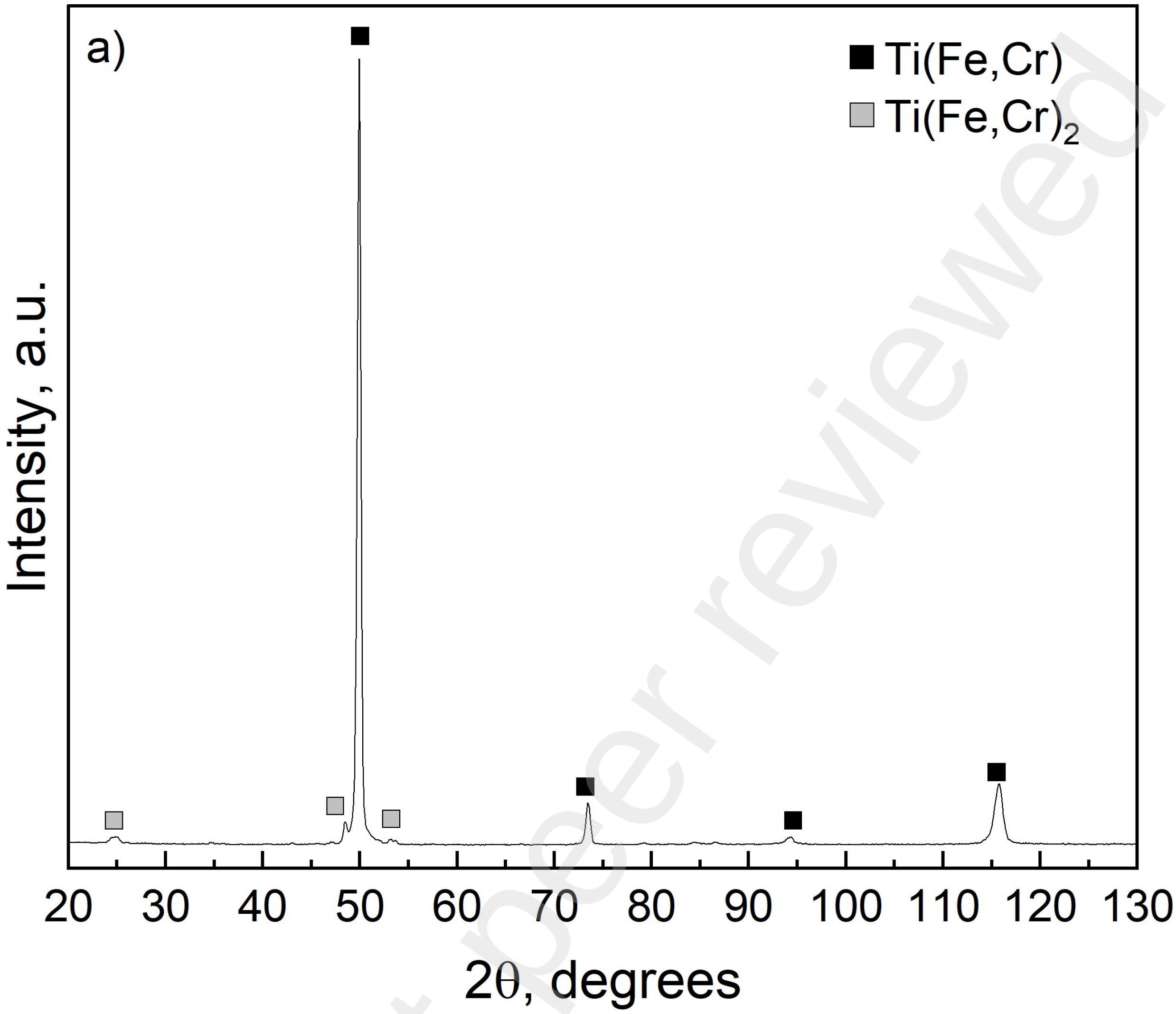

a)
Ti(Fe,Cr)
Ti(Fe,Cr)2
Intensity, a.u.
20
30
40
50
60
70
80
90
100
110
120
130
2θ, degrees

Fig 1a

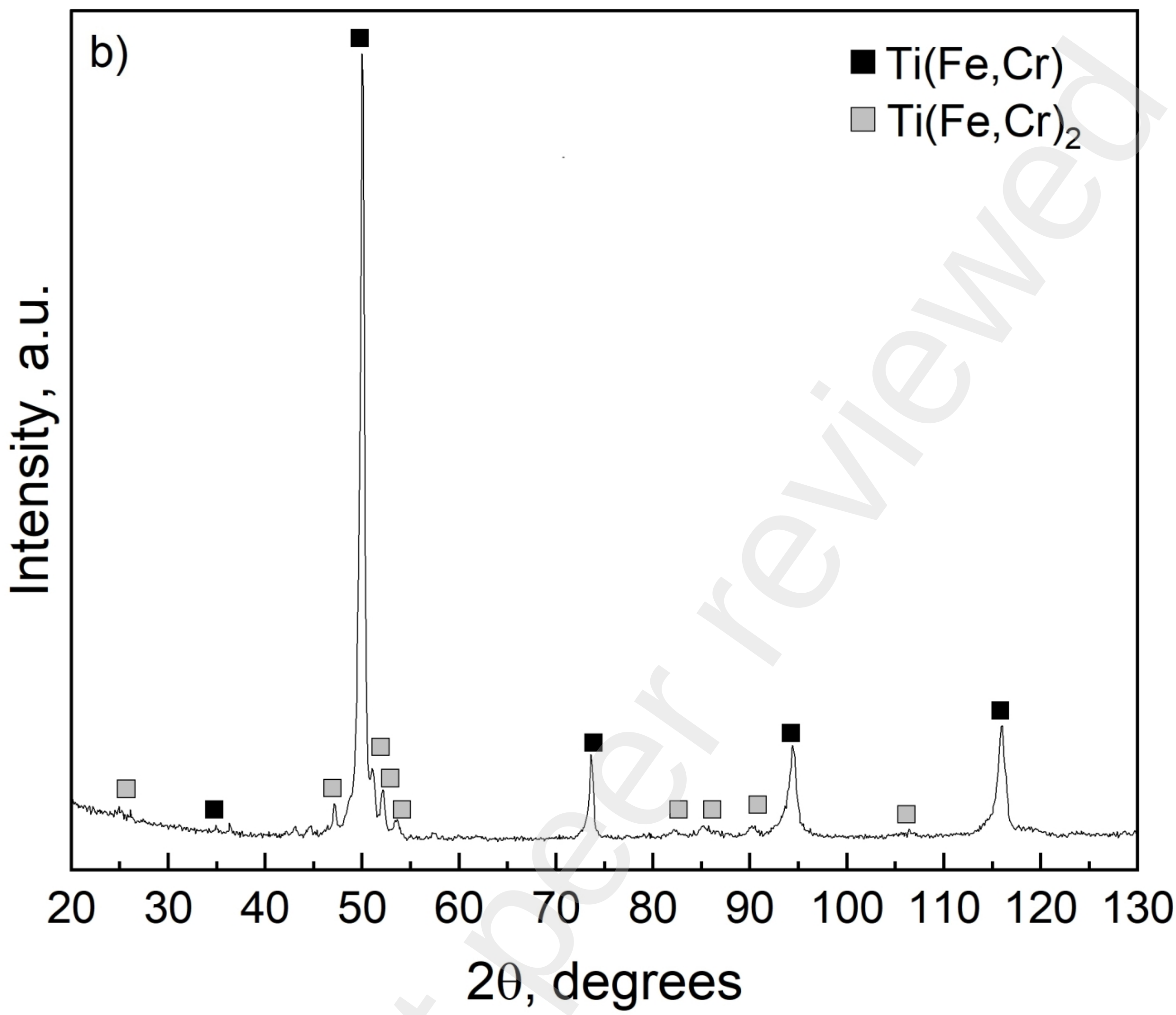

b)
Ti(Fe,Cr)
Ti(Fe,Cr)2
Intensity, a.u.
20
30
40
50
60
70
80
90
100
110
120
130
2θ, degrees

Fig 1b

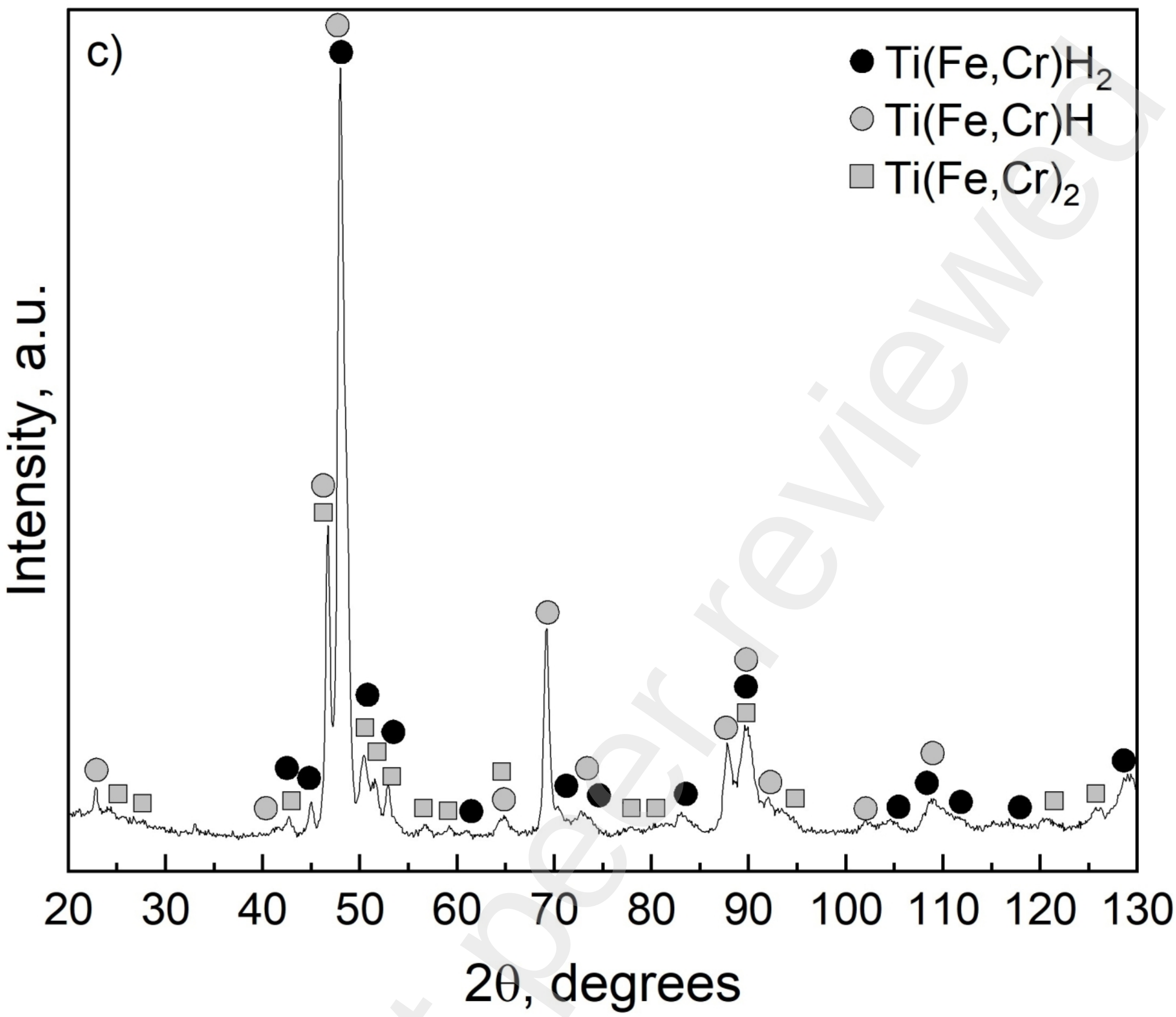

c)
Ti(Fe,Cr)H$_2$
Ti(Fe,Cr)H
Ti(Fe,Cr)$_2$
Intensity, a.u.
20
30
40
50
60
70
80
90
100
110
120
130
2θ, degrees

Fig 1c

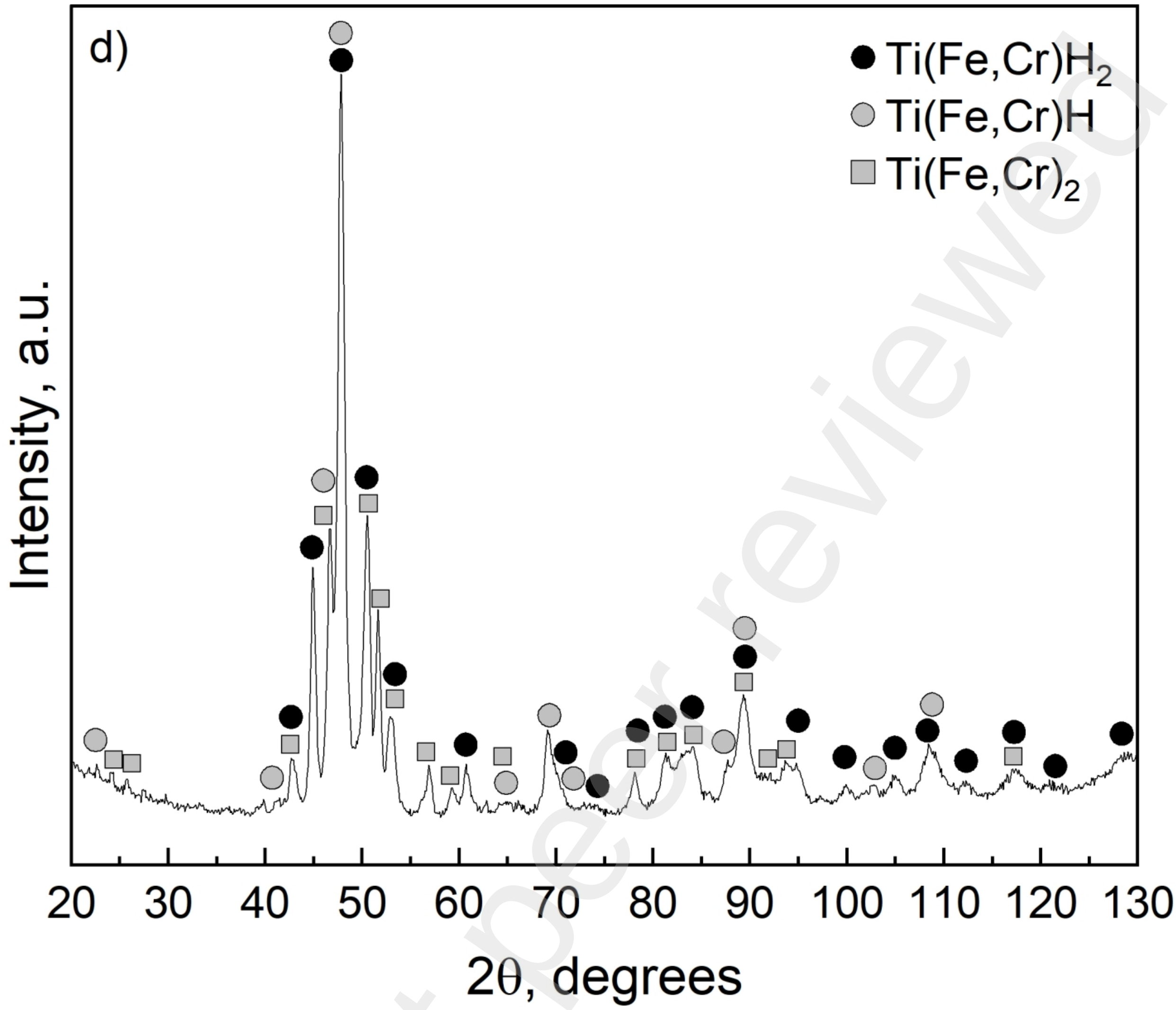

d)
Ti(Fe,Cr)H2
Ti(Fe,Cr)H
Ti(Fe,Cr)2
Intensity, a.u.
20
30
40
50
60
70
80
90
100
110
120
130
2θ, degrees

Fig 1d

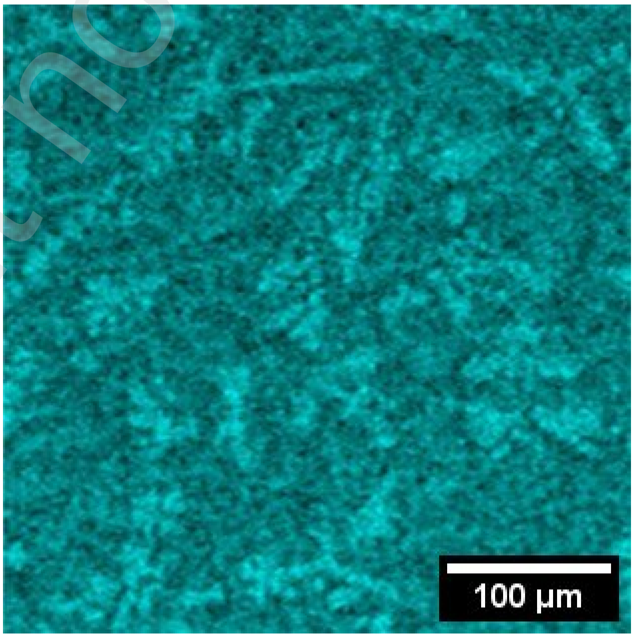

100 μm

Fig 2a

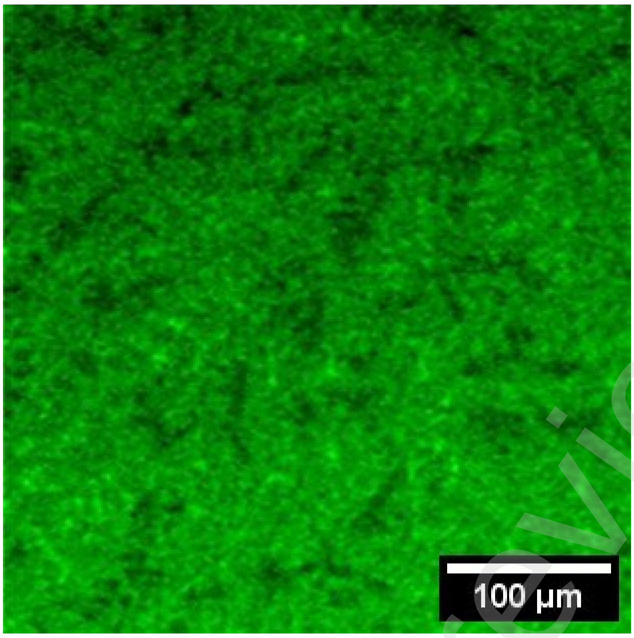


Fig 2b

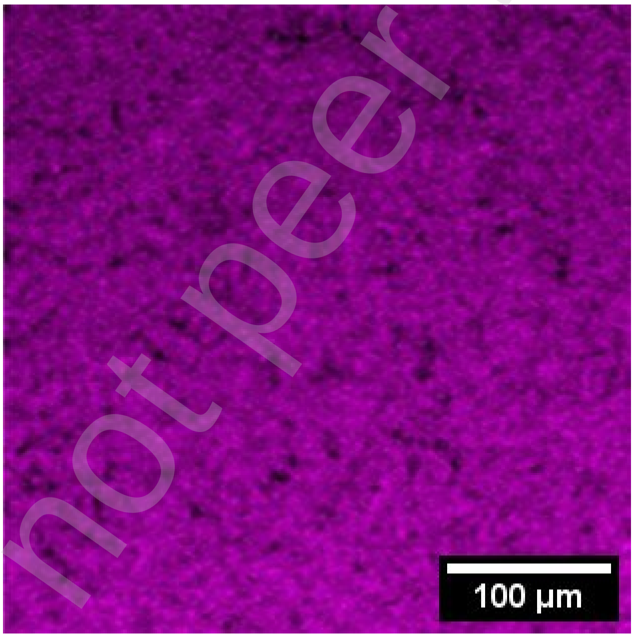


Fig 2c

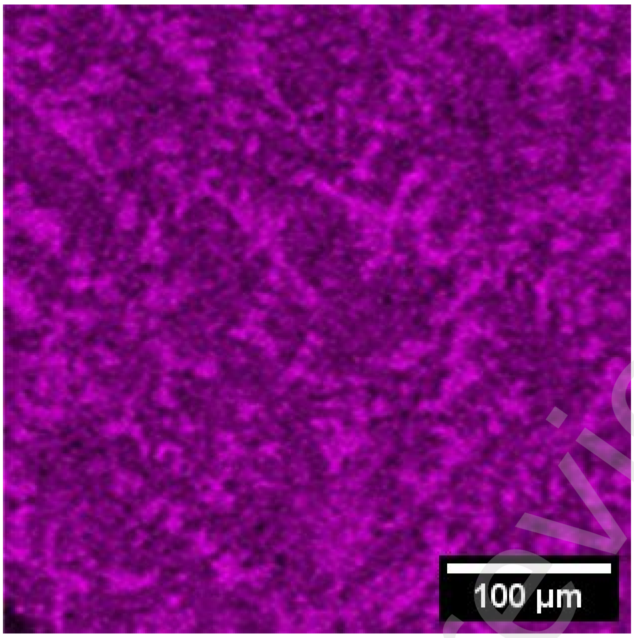
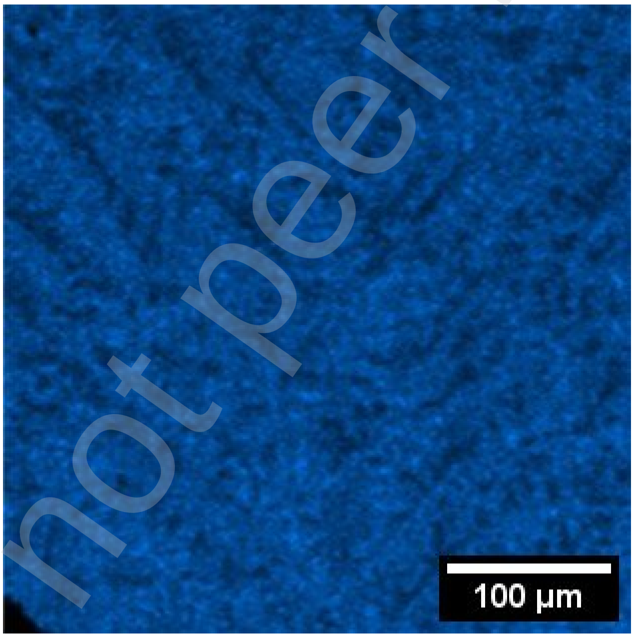


Fig 3a



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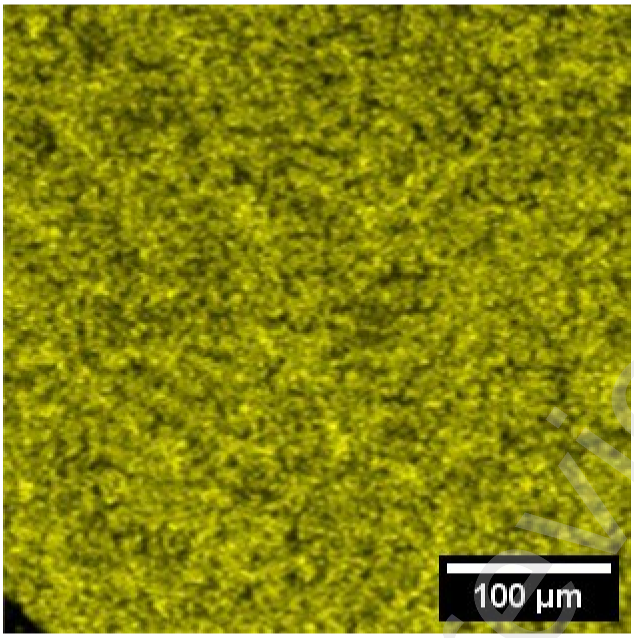

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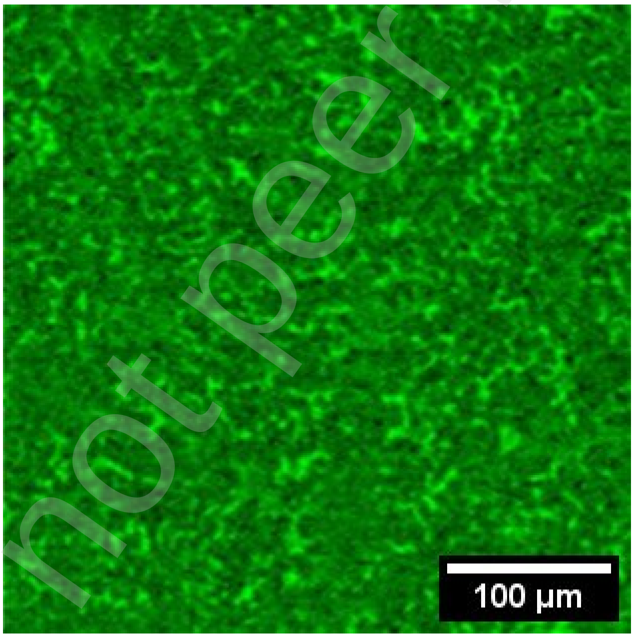

Fig 3d

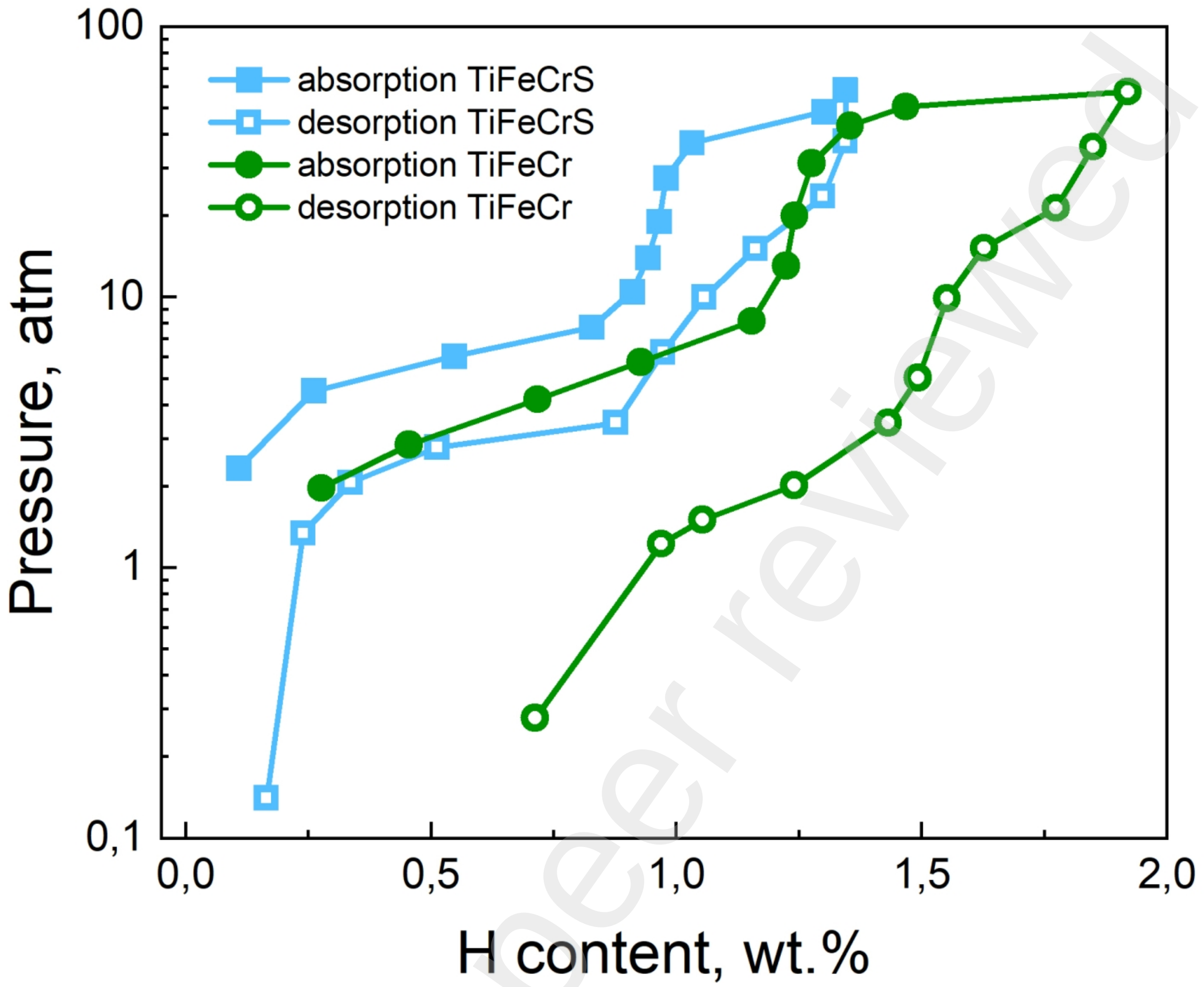
100
10
1
0,1
Pressure, atm
0,0
0,5
1,0
1,5
2,0
H content, wt.%
absorption TiFeCrS
desorption TiFeCrS
absorption TiFeCr
desorption TiFeCr

Fig. 4

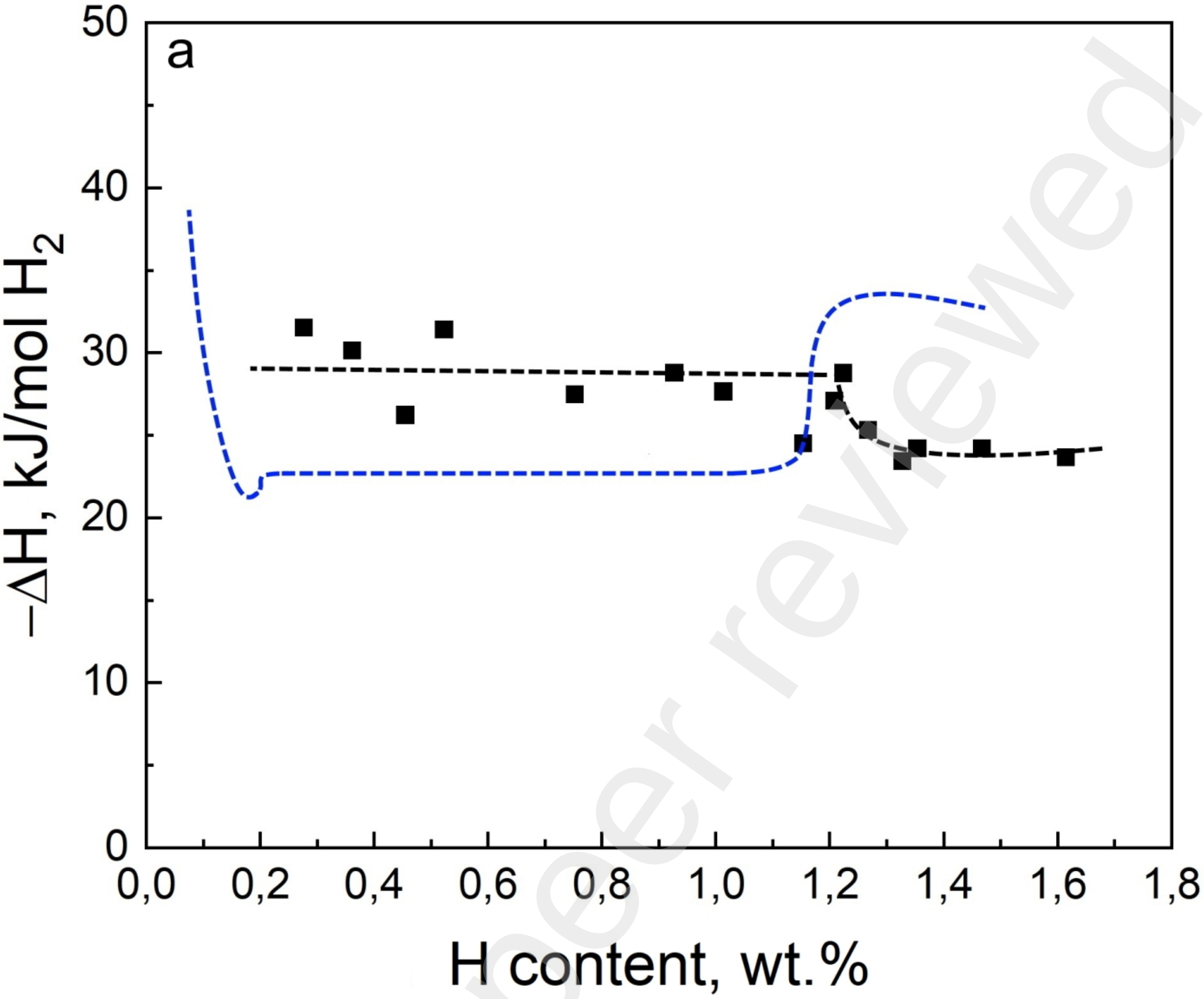

Fig. 5a TiFeCr abs

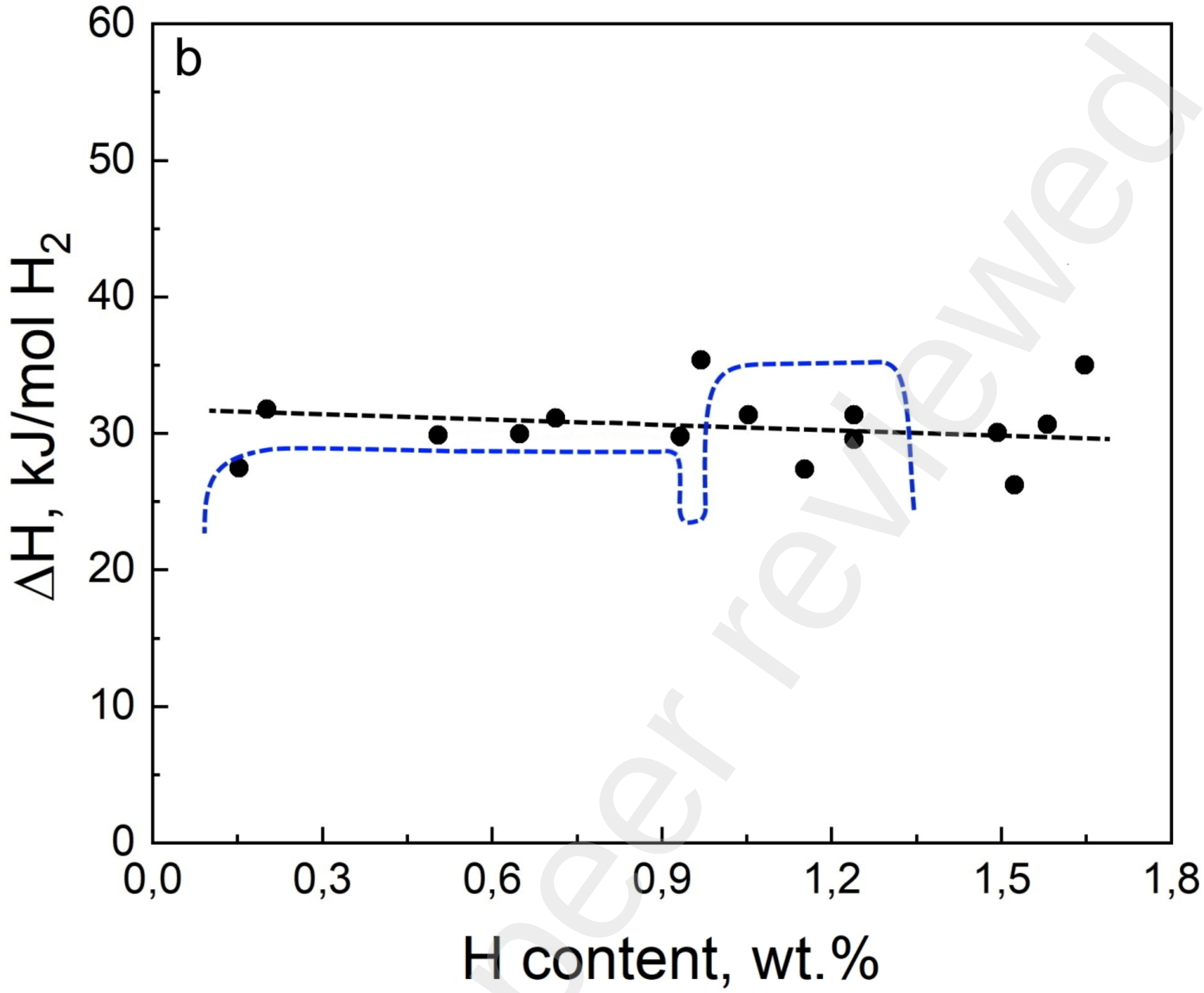

Fig. 5b TiFeCr des

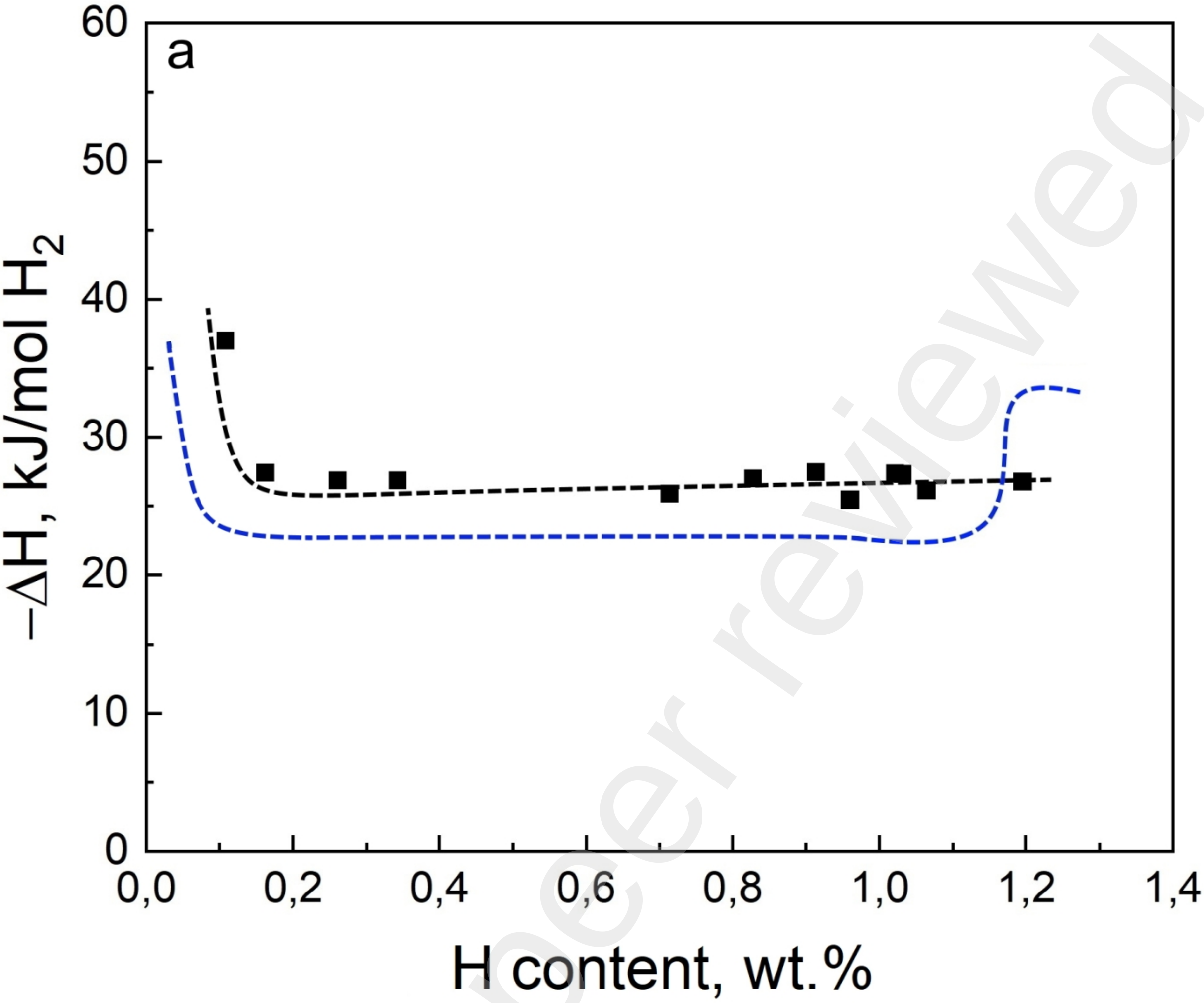

Fig 6a TiFeCrS abs

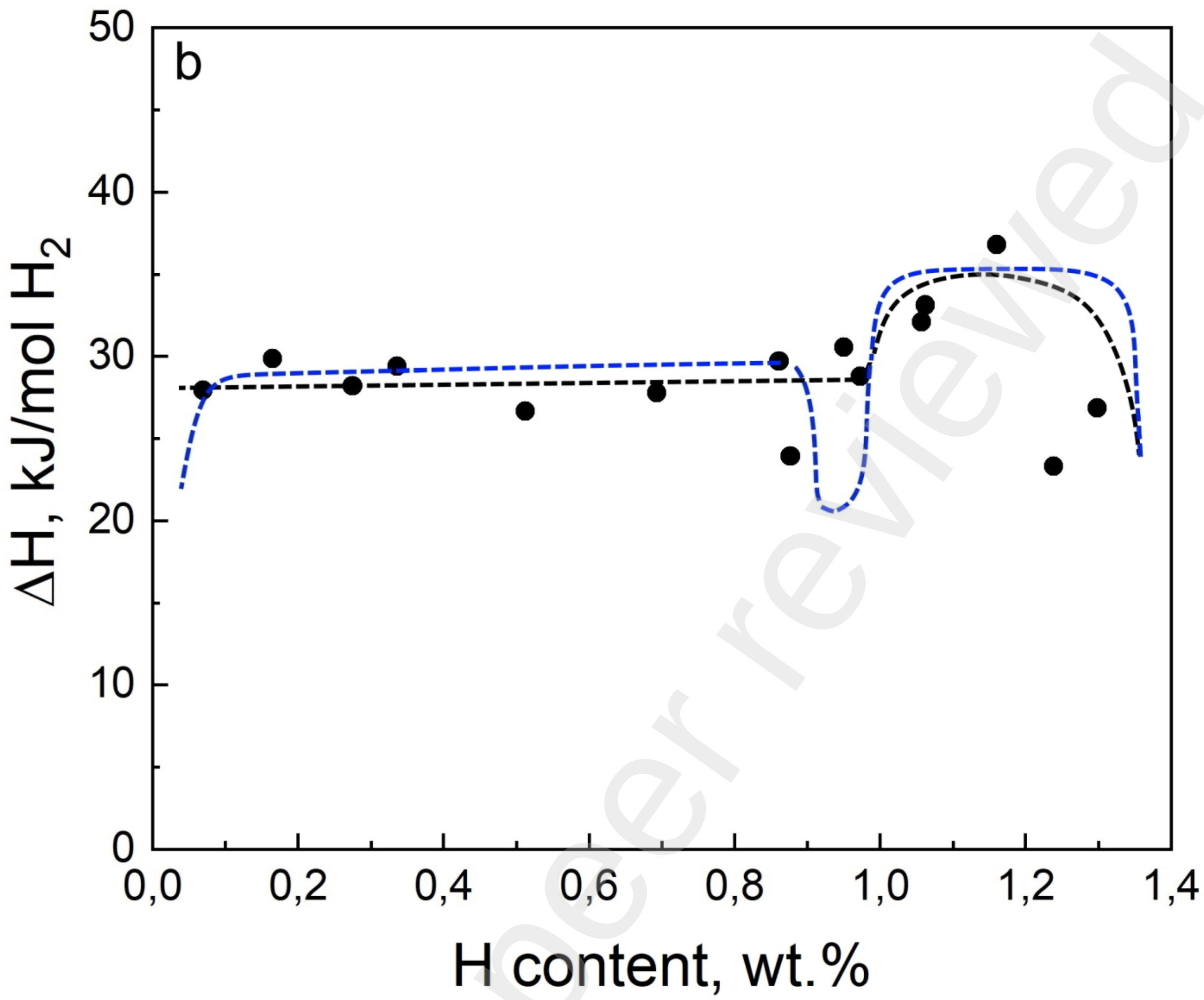

Fig 6b TiFeCrS des